

#FABCONSQLCON2026

FABCON

Microsoft Fabric
COMMUNITY CONFERENCE

SQLCON

Microsoft SQL
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Semantic Link Labs: The Data Detective's Toolkit

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The DAX Shepherd



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 Nashville, TN, USA

 Began Career as a SQL Server DBA

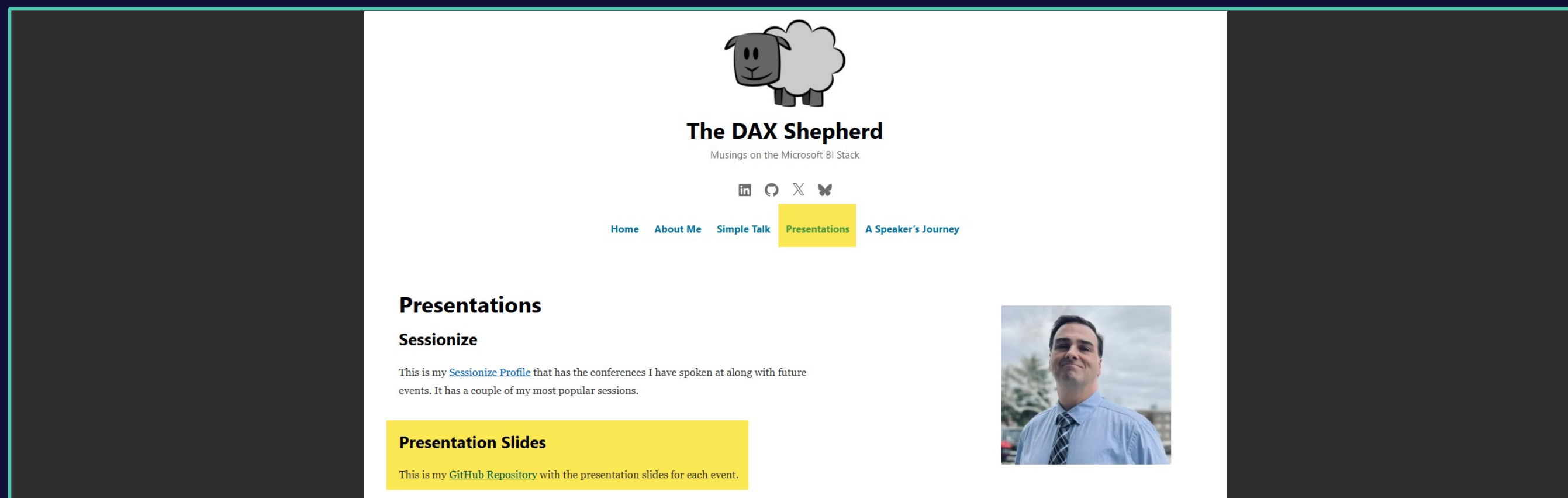
 Transitioned to Microsoft BI Stack

 Data Engineering to Data Modeling

DOWNLOAD THIS PRESENTATION

All slides, resources, and demo files available:

The DAX Shepherd
[www.thedaxshepherd.com/
presentations](http://www.thedaxshepherd.com/presentations)



Standing on the Shoulders of Giants



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Grateful for the people who share their knowledge, encourage us, answer questions, and help the rest of us learn faster.

The Case for Semantic

Why this matters and what Semantic Link Labs enables



OPEN THE CASE

YOU ARE HERE



CASE DEMOS



DETECTIVE AGENCY



CASE CLOSED



Sound Familiar?

**"It was fine on
Friday"**

Why BI Teams Need Detectives

often, BI teams investigate only after something breaks

- Model size creeps up
- Best practices drift over time
- Small changes leave a wide trail
- Performance issues appear before



cause

Symptoms aren't evidence — they're just the first clue.

Monk Moment: "Something's off"

That instinct is valuable when paired with evidence.

Use the detective habit:

- Notice the detail that doesn't fit
- Follow the trail of dependencies
- Prove it with evidence
- Turn investigation into prevention

What You'll Learn by the end, you'll be able to:

Investigate model performance with evidence
Trace dependencies and understand change impact
Detect breakage and hygiene issues proactively
Turn investigations into a reusable playbook

The Detective

Use this loop for every mystery you investigate.

Method

Investigation loop:

- Symptom (what users feel)
- Evidence (what we can prove)
- Suspects (ranked likely causes)
- Fix (targeted action)
- Prevention (make it repeatable)

Detective's Toolkit

SL

Semantic Link

Microsoft first-party library
Runs inside Fabric notebooks
Core: list tables, run DAX, read metadata

SLL

Semantic Link Labs

Open source
Extends Semantic Link
BPA · VertiPaq · report scanning

FN

Fabric Notebook

The runtime environment
Where detective work happens
Python + Spark, pre-installed libraries

Semantic Link Labs

Microsoft Fabric



Semantic Link

- List Tables
- List Workspaces
- List Models
- List Reports



Semantic Link Labs



Semantic Link + Semantic

Link Labs What this

unlocks:

- Inspect Fabric through code, not just the UI
- Investigate assets and metadata systematically
- Trace relationships and understand downstream impact
- Catch problems earlier and reduce guesswork
- Turn one-off investigations into repeatable practice

Semantic Link

Microsoft's first-party Fabric Python library

- Runs only inside the Fabric runtime
- Foundation layer: `import sempy.fabric as fabric`
- Preinstalled in the default runtime
 - Update with `%pip install -U semantic-link`
- Extended by Semantic Link Labs

Semantic Link Labs

- Michael Kovalsky
 - aka “Kovalsky’s Laboratory”
- “Expansion Pack” — Kurt Buhler
- Built on Semantic Link
 - `import sumpy.fabric as fabric`
- Open source on GitHub
 - Active Development

Installing in Microsoft Fabric

Update packages in a notebook

cell

Semantic Link

Preinstalled in the Fabric runtime

```
%pip install -U semantic-link
```

Semantic Link Labs

Install separately when needed

```
%pip install -U semantic-link-labs
```

Follow the Evidence

Real demos that uncover what is happening in Fabric



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Case File: The Unidentified

Model at the scene. Catalog everything before you touch it.

VICTIM: A semantic model in a new Fabric workspace

SYMPTOMS:

- No programmatic inventory — structure is trapped in the UI
- Relationships exist in the model but not in the data — or vice versa

Case File: The

Relationship Web

He sees connections others miss. So does find_relationships.

VICTIM: A semantic model with undetected relationship gaps

SYMPTOMS:

- Power BI shows only the relationships it was told about
- Data suggests more connections exist — but nobody modeled them

Case File: Running the

Numbers
Your DAX knowledge is trapped in the desktop. Time to set it free.

VICTIM: DAX knowledge locked inside Power BI Desktop

SYMPTOMS:

- Can't query the model programmatically from a pipeline
- DAX results can't feed downstream Python or notebook logic
- No way to automate recurring DAX calculations

MO: The Detective's Toolkit (Cases 1-3)

DEMO

Let's go investigating...

Case Board: Investigations 1-3

Cases closed. Evidence on record.

CASE 01

The Unidentified Model

SOLVED

CASE 02

The Relationship Web

SOLVED

CASE 03

Running the Numbers

SOLVED

Case File: Something Doesn't

Add Up

Every relationship has two sides. Not every key agrees.

VICTIM: Referential integrity violations hiding in your model

SYMPTOMS:

- Fact table keys that point to dimension values that don't exist
- Blank members appearing in slicers or visuals unexpectedly
- Aggregations silently drop rows with no error

Case File: The Evidence Isn't Ready Yet

The witness statement exists. Monk knows it. But it's not in the file

VICTIM: A Delta table just written to OneLake

SYMPTOMS:

- Table confirmed written in the notebook
- Not visible in the SQL Analytics Endpoint

Case File: The Witness

The witness was there. Then they weren't.
Vanished

VICTIM: A Delta table deleted from the Lakehouse

SYMPTOMS:

- Table confirmed visible in SQL Analytics Endpoint
- Then gone — query returns nothing
- No backup process, no pipeline to re-run

EMO: The Detective's Toolkit (Cases 4-6)

DEMO

TIME

Let's work the case.....

Case Board: Investigations 4-6

Cases closed. Evidence on record.

CASE 04

**Something
Doesn't Add Up**

SOLVED

CASE 05

**The Evidence
Isn't Ready Yet**

SOLVED

CASE 06

**The Witness
Vanished**

SOLVED

Case File: The Workspace-Wide Sweep

One detective can only cover so much ground.

VICTIM: An entire Fabric workspace — all semantic models

SYMPTOMS:

- BPA run on one model — what about the rest?
- Governance gaps accumulate silently across the workspace
- No historical trend data for model quality

DEMO: The Inspection (Best Practice Analyzer)

DEMO

TIME

There's only one way to know.....

Case File: The Report That

Lied

The report is lying to you. I'm not sure how yet. But I will be.

VICTIM: A Power BI report connected to a semantic model

SYMPTOMS:

- Visuals reference fields that were renamed or removed
- Report-level measures live in the report instead of the semantic model

DEMO: The Report Is Lying (Report Analysis)

DEMO

TIME

Let's see what doesn't add up.....

Case File: The

Start with the model:
Profile

Find out who's taking up space before they become a problem

VICTIM: The SF Case Files semantic model

SYMPTOMS:

- Memory usage is hidden behind the scenes
- Large tables and high-cardinality columns are hard to spot by eye
- Model bloat slows performance and makes optimization guesswork

DEMO: The Profile (VertiPaq Analyzer)

DEMO

TIME
This deserves a closer look...

Build the Detective Agency

Monitoring, automation, and repeatable practices for the long haul



OPEN THE CASE



CASE DEMOS



DETECTIVE AGENCY



CASE CLOSED

YOU ARE HERE



From Detective to Detective Agency

1

**Ad-Hoc
Investigations**

2

**Reusable
Notebooks**

3

**Standard
Health Checks**

BPA + VertiPaq
+ dependencies

4

**Run Pre/Post
Change**

Catch regressions
before users do

5

**Share &
Build Playbook**

Your team runs
it without you

The Precinct Goes

Automated. One function call. One dashboard. A repeatable monitoring view.

WHAT THIS DOES:

- Creates a pre-built workspace monitoring dashboard
- Scopes it to the current workspace
- Gives the team a repeatable monitoring starting point

WHY IT MATTERS:

- Moves from ad-hoc investigation to operational visibility
- Makes monitoring easier to share across the team
- Helps confirm issues faster without rebuilding the view manually

PRE-REQUISITE:

- Workspace monitoring must be enabled first

DEMO: The Precinct Goes Automated (Workspace Monitoring Dashboard)

DEMO

TIME

The dashboard will confirm it....

Case

Key takeaways, next steps, and where to start with Semantic Link L



OPEN THE CASE



CASE DEMOS



DETECTIVE AGENCY



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Case Closed: Key Takeaways

1

Evidence First

Turn “something’s off”
into
evidence using
Semantic Link Labs

2

Map the Blast Radius

Use dependencies to
understand
change impact before
you make it

3

Build the Playbook

Make investigations
repeatable
so you find issues
before users do

More from the Crime Lab

*SLL ships ready-to-run notebooks for common Fabric governance tasks
— each one a solved case waiting to open*

Best Practice Analyzer

— Run BPA rules across models, export results, and spin up a Direct

Model Optimization

— Fix column encoding, remove auto date tables, clean up model

Migrate to Direct Lake

— Convert import-mode semantic models to Direct Lake in one

Workspace Monitoring

— Provision a KQL Eventhouse and real-time dashboard with a single

Report Rebind

— Retarget reports to a new semantic model across an entire
workspace

github.com/microsoft/semantic-link-labs → [notebooks/](#)

Next Steps

- Run BPA + VertiPaq + dependency scan on one of your own models this week
- Create a case notebook template your team can run without you
- Schedule `run_model_bpa_bulk()` — make governance automatic

Resources

Semantic Link

<https://learn.microsoft.com/en-us/fabric/data-science/semantic-link-overview>

Semantic Link Labs

github.com/microsoft/semantic-link-labs

Session materials (slides + notebooks)

github.com/thedaxshepherd/Presentations

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our Fabric product group and
researchers.

<https://aka.ms/JoinFabricUserPanel>

Join the SQL User Panel



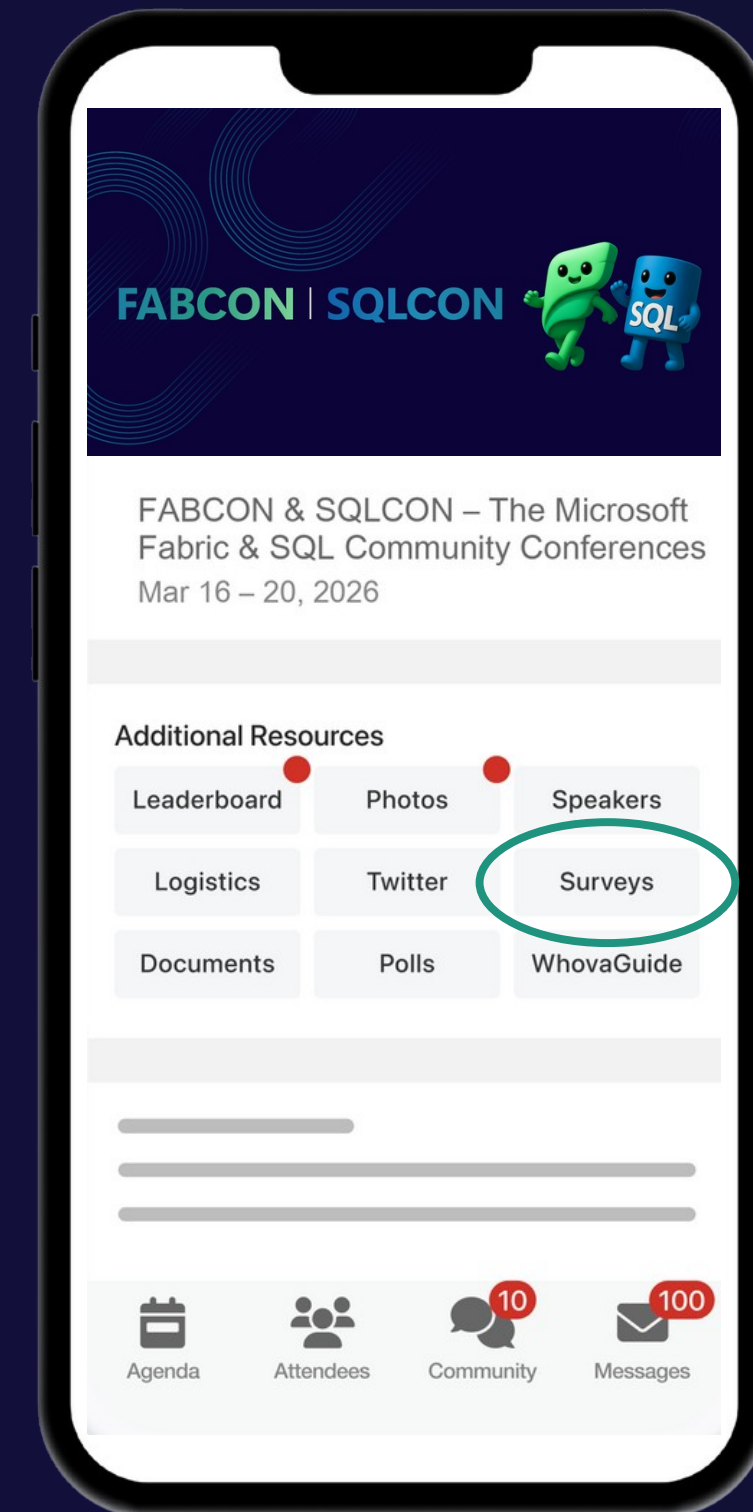
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needs

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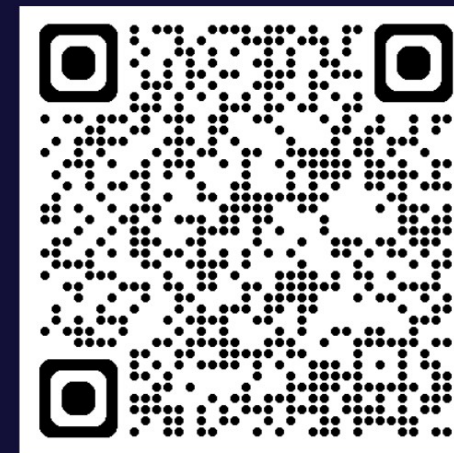


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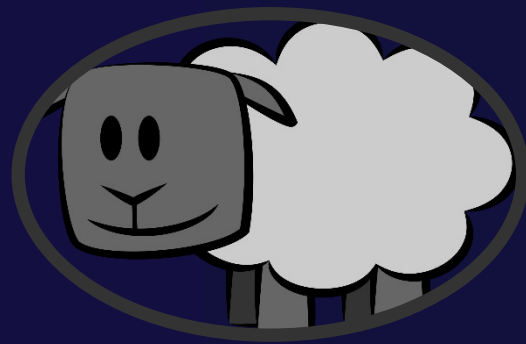
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THANK YOU



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